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TTMI Helps Customer Fleets Reduce Carbon Emissions and Operational Costs Using AWS

TTMI built a fleet management solution entirely on AWS to deploy quickly at scale and helps its customers lower carbon emissions, vehicle issues, and operations costs.

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Scaled solution

to meet customer needs

Achieved rapid deployment



Up to 40%

reduction in customer carbon emissions

60%

reduction in customer vehicle issues

10% –17%

reduction in customers' operational costs

Overview

In 2019, [Toyota Tsusho Mobility Informatics Pte. Ltd.](#) (TTMI) was established through a collaboration of three Toyota Group members to create cutting-edge technologies and deliver state-of-the-art connected vehicle services to global fleet owners. For these ambitious aims, TTMI needed to build a rapidly scalable, cloud-based data collection and fleet management solution to adapt to its customers' dynamic needs.

In 2021, TTMI rolled out Fleemo, a comprehensive fleet management system that marries logistics seamlessly. By using the scalable features of Amazon Web Services (AWS), TTMI could cater to a variety of customer needs spanning driver analytics to traffic routing. Furthermore, TTMI used the rapid deployment capability of AWS to efficiently meet regulatory stipulations in new countries, aiding their customers in reducing carbon emissions, operation costs, and vehicle issues.



Opportunity | Creating a Fleet Management Solution on AWS

TTMI, a joint establishment of Toyota Tsusho Corporation, Toyota Tsusho Asia Pacific, and Toyota Tsusho NEXTY Electronics (Thailand), commenced operations in November 2019. The company, headquartered in Singapore, with a Thailand office established in 2020, launched services in Singapore, Papua New Guinea, Cambodia, and Thailand in 2021, later expanding to Japan, the United Arab Emirates, and Fiji. TTMI offers its connected services to its customers, each operating a fleet of 10-100 plus connected vehicles.



To cater to varying customer requirements, some needing real-time updates while others relying on periodic reports, TTMI constructed a data collection and fleet management solution using AWS. TTMI opted for AWS after assessing various cloud providers, seeking a service that offered agility for swift deployment and global expansion, while also adhering to region-specific regulations. “We were able to get into the marketplace fast using AWS,” says Juan Ramirez, software architect at TTMI. “There was no learning curve, and AWS had the expertise and resources we were looking for.”

Solution | Innovating on AWS

While building Fleemo, TTMI used AWS as it aimed to resolve complex issues and comply with Toyota Group’s strict data collection guidelines. Incorporating firewalls and maintaining isolated infrastructure was part of this process. “Using AWS, we can comply with the security guidelines we need to meet,” says Ramirez. “We implement the out-of-the-box monitoring tools from AWS services, and Amazon MSK has been our cornerstone.”

As TTMI expands into the Middle East and Europe, regulatory compliance and product availability are paramount. Using AWS, TTMI is accomplishing secure, rapid deployment with high availability across regions. “The technology we use on AWS is very reliable,” says Richard Lee, general manager at TTMI. “We are never inhibited from a business perspective on what we can achieve.”

TTMI handles vast amounts of streaming information online, such as weather conditions, driver capabilities, and future traffic conditions. This data is processed using statistical models. Using AWS, TTMI is building the data infrastructure to analyze the information it collects, with the aim of further improving customer experiences. “Our route-planning software affects travel time and traffic conditions for fleet drivers in a city,” says Ramirez. “Being able to provide a solution that can improve this for our customers using AWS has been a great experience.”

Outcome | Reducing Customer Costs, Vehicle Issues, and Carbon Footprint

Fleemo is built on an innovative architecture composed of numerous microservices. These components collect and process telematics data from devices installed within client vehicles. This data, collected continuously



connected fleet, is securely streamed through [Amazon Managed Streaming for Apache Kafka](#) (Amazon MSK), which securely streams data in near real time with a fully managed, highly available Apache Kafka service. Using Amazon MSK, TTMI achieves an uninterrupted data flow.

The data, once processed using Amazon MSK, takes one of two routes. Either it is relayed to a data lake, where it's analyzed for insights, or it is directly integrated into TTMI's in-house database infrastructure. This dual-path processing system allows for comprehensive use of data, making sure all information is used to its fullest potential in the pursuit of fleet management optimization.

The application of microservices has allowed TTMI to offer unique features for individual customers without disrupting the system's functionality. In cases where customers require dedicated infrastructure and customized features in addition to the existing software-as-a-service solution, TTMI can provide a new infrastructure for them using AWS and DevOps technologies. "The architecture we built on AWS is uncomplicated in nature but very powerful," says Ramirez. "We can grow across the board because we have the differentiators to meet customer needs on an individual basis."

The benefits of implementing Fleemo have been significant. Customers globally have reported up to 40 percent improvement in carbon reduction, partly attributed to reduced vehicle idling times. The incorporation of Fleemo has also led to a 60 percent reduction in recorded vehicle issues, credited to predictive maintenance alerts, driver behavior analytics, optimized delivery routes, and live traffic and weather updates. Operational costs have been cut by 10–17 percent in some countries, thanks to improved fuel efficiency, optimized delivery routes, and reduced vehicle idling times.

Learn More

To learn more, visit aws.amazon.com/msk/.

About Toyota Tsusho Mobility Informatics Pte. Ltd.

Toyota Tsusho Mobility Informatics Pte. Ltd. was jointly established in 2019 by three businesses in the Toyota Group. The company has developed advanced technologies to provide connected mobility services for customers such as ride-hailing services.

AWS Services Used

Amazon Managed Streaming for Apache Kafka (Amazon MSK)

Amazon MSK makes it easy to ingest and process streaming data in real time with fully managed Apache Kafka.

[Learn more »](#)



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